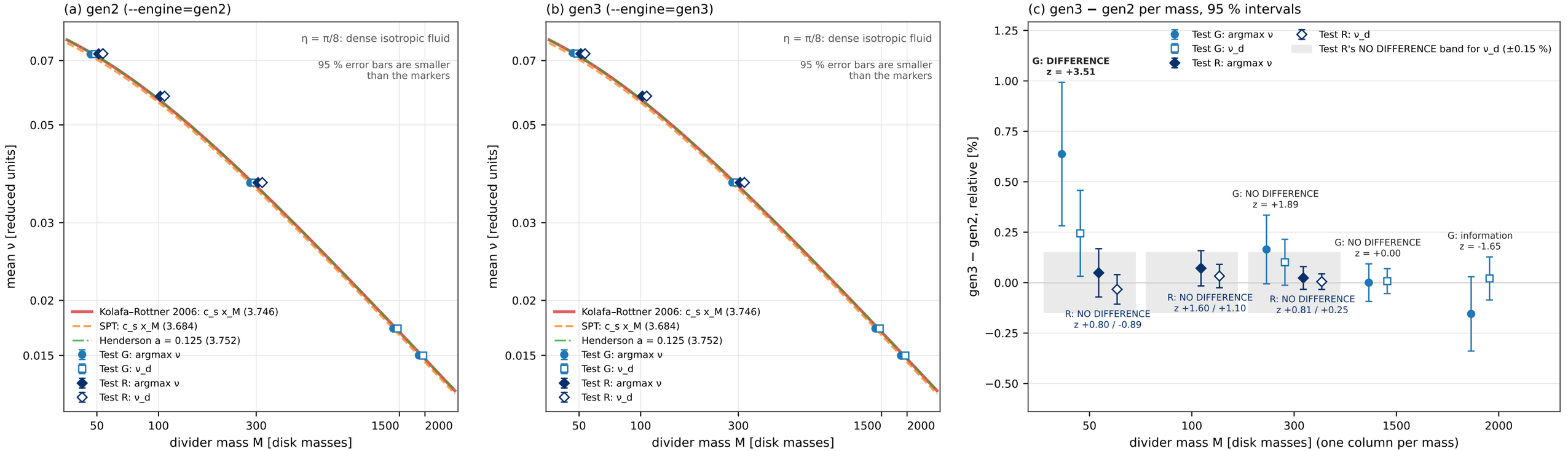
# Test G: FAIL at M = 50; replication Test R: ENGINES AGREE

N = 100 disks (50 per compartment), radius 0.5, L0 = 10, H = 10, η = π/8; one KOA binary, engine-gen3 a643bc7 (target koa), run as gen2 (a) and gen3 (b); the same seeds in both arms. Test G: 400 trajectories per engine and mass (M = 2000: 100); Test R: 3600




Cell epi8_H_H10_L10: N = 100, N_s = 50, disk radius 0.5, L0 = 10, H = 10, wall thickness 0.05, eta = pi/8. Test G's rule (sec. 4.7.29): NO DIFFERENCE if |z| < 2, DIFFERENCE if |z| >= 3, argmax nu, at M = 50, 300, 1500 (M = 2000 information). Test R's rule (decision 28): per mass DIFFERENCE CONFIRMED if |z| >= 3.5 by either estimator, NO DIFFERENCE if both |z| < 3.5 and nu_d's 95 % interval lies within +-0.15 %.
Error bars 95 %: (a, b) mean +- 1.96 SE; (c) Test G: 100 (d -+ 1.96 s) / mean_gen2 (its verdict); Test R: d -+ 1.96 SE(d), delta method (its verdict). nu_d: the sec. 4.4.13 item 4a fit per trajectory (Test G: information; Test R: registered). Values reproduce both verdicts' printed tables (secs. 4.7.54, 4.7.66). Test G and Test R pooled (Report R1): M = 50, nu_arg +0.107 +- 0.058 % (z 1.86), nu_d -0.006 +- 0.035 %.
Theory lines: nu = c_s(eta) x_M, c_s adiabatic (kT = m = 1) at eta = pi/8; x_M = K(alpha)/(2 pi L_eff), L_eff = 8.975 (tests_20260913.x_of).
tables: hspist3/experiments_m1_figures_261011/tables/ (main 752d5754); script: hspist3/experiments_m1_figures_261011/m1_figures_v2_261011.py (main 204c322d)